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INSTITUTE

The diagram consists of a rectangular box at the top with a folded top-left corner. Inside this box is the word 'INSTITUTE' in a bold, black, serif font. Below the box is a large, hollow downward-pointing arrow. At the base of the arrow is a horizontal banner with wavy, torn edges. Inside the banner are the words 'VISION' and 'MISSION' in the same bold, black, serif font, separated by a wide space.

VISION MISSION

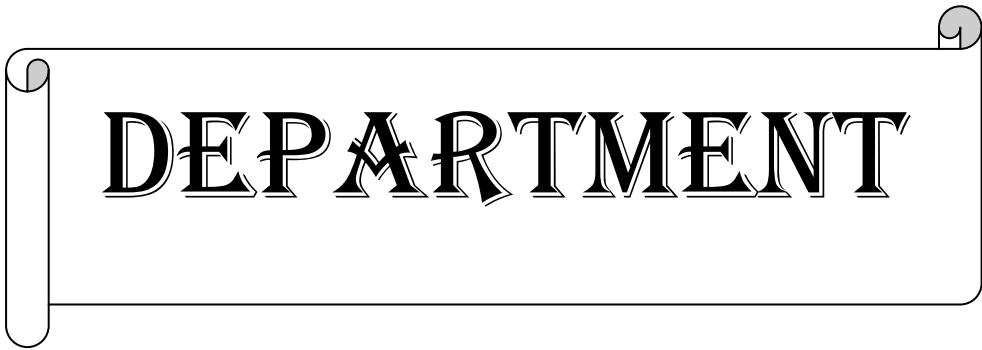
INSTITUTE VISION AND MISSION

VISION

To be a premier technological institute striving for excellence with global perspective and commitment to the nation.

MISSION

- To produce engineering graduates of professional quality and global perspective through Learner Centric Education.
- To establish linkages with government, industry and research laboratories to promote R&D activities and to disseminate innovations.
- To create an eco-system in the institute that leads to holistic development and ability for life-long learning..



DEPARTMENT



VISION MISSION



DEPARTMENT VISION AND MISSION

Vision:

- To evolve as a centre of academic and research excellence in the area of Computer Science and Engineering.

Mission :

- To utilize innovative learning methods for academic improvement.
- To encourage higher studies and research to meet the futuristic requirements of Computer Science and Engineering.
- To inculcate Ethics and Human values for developing students with good character

**PROGRAM
EDUCATIONAL
OBJECTIVES,
PROGRAM OUTCOMES
& PROGRAM
SPECIFIC
OUTCOMES**

Program Educational Objectives (PEOs)

Graduates of this programme will :

PEO 1: Adapt to evolving technology.

PEO 2: Provide optimal solutions to real time problems.

PEO 3: Demonstrate his/her abilities to support service activities with due consideration for Professional and Ethical Values.

Programme Specific Outcomes (PSO s):

A graduate of the Computer Science and Engineering Program will be able to:

PSO 1: Use Mathematical Abstractions and Algorithmic Design along with Open Source Programming tools to solve complexities involved in Programming. [K3]

PSO 2: Use Professional engineering practices and strategies for development and maintenance of software. [K3]

Program Outcomes (POs):

Computer Science Engineering Graduates will be able to:

1. **Engineering knowledge:** Apply the knowledge of Mathematics, Science, Engineering Fundamentals and Concepts of Computer Science Engineering to the solution of complex Engineering problems. **[K3]**
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of Mathematics, Natural Sciences and Computer Science. **[K4]**
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specific needs with appropriate consideration for the public health and safety, and the cultural, societal and environmental considerations. **[K5]**
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions. **[K5]**
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex Engineering activities with an understanding of the limitations. **[K3]**
6. **The Engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional Engineering practice. **[K3]**
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development. **[K3]**
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the Engineering practice. **[K3]**
9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings. **[K6]**
10. **Communication:** Communicate effectively on complex Engineering activities with the Engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions. **[K2]**
11. **Project management and finance:** Demonstrate knowledge and understanding of the Engineering and Management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments. **[K6]**
12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change. **[K1]**

ACADEMIC CALENDAR

A.Y. 2019-20

B.TECH III SEM			
DESCRIPTION	FROM	TO	WEEKS
COMMENCEMENT OF CLASS WORK	10.06.2019		
I UNIT OF INSTRUCTIONS	10.06.2019	03.08.2019	8 WEEKS
I MID EXAMINATION	05.08.2019	10.08.2019	1 WEEK
II UNIT OF INSTRUCTIONS	12.08.2019	05.10.2019	8 WEEKS
II MID EXAMINATION	07.10.2019	12.10.2019	1 WEEK
COMPREHENSIVE TEST	14.10.2019	19.10.2019	1 WEEK
PREPARATION AND PRACTICALS	21.10.2019	25.10.2019	5 DAYS
END EXAMINATIONS	28.10.2019	09.11.2019	2 WEEKS
B.TECH IV SEM			
COMMENCEMENT OF CLASS WORK	11.11.2019		
I UNIT OF INSTRUCTIONS	11.11.2019	04.01.2020	8 WEEKS
I MID EXAMINATION	06.01.2020	11.01.2020	1 WEEK
II UNIT OF INSTRUCTIONS	13.01.2020	07.03.2020	8 WEEKS
II MID EXAMINATION	09.03.2020	14.03.2020	1 WEEK
COMPREHENSIVE TEST	16.03.2020	21.03.2020	1 WEEK
PREPARATION AND PRACTICALS	23.03.2020	27.03.2020	5 DAYS
END EXAMINATIONS	30.03.2020	11.04.2020	2 WEEKS

CO-CURRICULAR AND EXTRA CURRICULAR ACTIVITIES

05/06/2019 – World Environment Day	15/09/2019 – Engineers Day		21/03/2019 – International Forest Day
21/08/2019 to 24/08/2019– 1st year B.Tech. Introduction program	16/09/2019 – World Ozone Day	22/12/2019 – National Mathematics Day	22/03/2019 –World Water Day
15/08/2018 – Independence Day	September – Intramurals	26/01/2019 – Republic Day	23/03/2019 – World Meteorological Day
05/09/2019 – Teachers Day	11/11/2019 – National Education Day	08/03/2019 – International Women’s Day	24/03/2019 – Earth Hour
	03/12/2019 – Antipollution Day		In the month of March – Association Days



SRI VASAVI ENGINEERING COLLEGE (Autonomous)

Pedatadepalli, TADEPALLIGUDEM-534 101, W.G. Dist.

Department Of Computer Science & Engineering (Accredited by NBA)



CLASS CONSOLIDATED TIME TABLE

Class: B.TechIVSemester

w.e.f : 18-11-2019

Sec : A

Class Coordinator : **Mrs. B. SriRamya**

Room No : B-303

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7
Time Day	(09.30 AM- 10.30 AM)	(10.30 AM- 11.20 AM)	(11.20 AM- 12.10 PM)	(12.10 PM- 01.00 PM)		(02.00 PM- 02.50 PM)	(02.50 PM- 03.40 PM)	(03.40 PM- 04.30 PM)
Mon	FLAT	JAVA	PCS-II		LUNCH BREAK	SE	CO	COI
Tue	PP	MEFA	COI	JAVA		SE	CO	PCS
Wed	JAVA	PP LAB				FLAT	PP	MEFA
Thu	SE	CO	JAVA	MEFA		PP	FLAT	TS-II
Fri	MEFA	SE	CO	JAVA		JAVA LAB		
Sat	FLAT	TS-II				FLAT	PP	LIB/SPORT S

Class: B.Tech IV Semester

w.e.f : 18-11-2019

Sec : B

Class Coordinator : **Mrs.MAnantha Lakshmi**

Room No : B-304

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7
Time Day	(09.30 AM- 10.30 AM)	(10.30 AM- 11.20 AM)	(11.20 AM- 12.10 PM)	(12.10 PM- 01.00 PM)		(02.00 PM- 02.50 PM)	(02.50 PM- 03.40 PM)	(03.40 PM- 04.30 PM)
Mon	PP	COI	FLAT	TS-II	LUNCH BREAK	JAVA LAB		
Tue	SE	JAVA	FLAT	CO		PCS-II		COI
Wed	FLAT	PP	JAVA	MEFA		SE	CO	PCS
Thu	JAVA	PP LAB				MEFA	PP	FLAT
Fri	CO	JAVA	MEFA	SE		TS-II		
Sat	SE	CO	PP	FLAT		JAVA	MEFA	LIB/SPORT S

Class: B.Tech IV Semester

w.e.f : 18-11-2019

Sec : C

Class Coordinator : **Mr.K Venkatesh**

Room No : B-202

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7
Time Day	(09.30 AM- 10.30 AM)	(10.30 AM- 11.20 AM)	(11.20 AM- 12.10 PM)	(12.10 PM- 01.00 PM)		(02.00 PM- 02.50 PM)	(02.50 PM- 03.40 PM)	(03.40 PM- 04.30 PM)
Mon	PP	CO	SE	MEFA	LUNCH BREAK	COI	JAVA	PCS
Tue	FLAT	TS-II	JAVA	PP		JAVA LAB		
Wed	JAVA	MEFA	CO	FLAT		SE	PP	FLAT
Thu	MEFA	CO	FLAT	JAVA		TS-II		
Fri	SE	PP LAB				MEFA	JAVA	COI
Sat	CO	SE	PCS-II			FLAT	PP	LIB/SPORTS

Class: B.Tech IV Semester

w.e.f : 18-11-2019

Sec : D

Class Coordinator : **Mrs.A. Sirisha**

Room No : B-203

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7
Time Day	(09.30 AM- 10.30 AM)	(10.30 AM- 11.20 AM)	(11.20 AM- 12.10 PM)	(12.10 PM- 01.00 PM)		(02.00 PM- 02.50 PM)	(02.50 PM- 03.40 PM)	(03.40 PM- 04.30 PM)
Mon	SE	JAVA	FLAT	PP	LUNCH BREAK	TS-II		
Tue	CO	PP LAB				SE	FLAT	PP
Wed	FLAT	MEFA	SE	PP		JAVA	COI	COI
Thu	MEFA	FLAT	TS-II	CO		JAVA LAB		
Fri	PP	CO	SE	JAVA		PCS-II		MEFA
Sat	JAVA	MEFA	JAVA	CO		PCS	FLAT	LIB/SPORTS

STAFF DETAILS:

S. No.	Course Code	Course Name	Section			
			A	B	C	D
1.	V18CST05	CO	Mr. B. Madhava Rao	Mr. B. Madhava Rao	Dr. S. P. Malarvizhi	Dr. S. P. Malarvizhi
2.	V18CST06	SE	Mr. M Vamsi Krishna	Mr. M Vamsi Krishna	Mr.K Venkatesh	Mr. K Venkatesh
3.	V18CST07	FLAT	Mr.M. Satyanarayana	Mrs. M Anantha Lakshmi	Mrs. M Anantha Lakshmi	Mr. P Srihari
4.	V18CST08	JP	Mrs. B.SriRamya	Mrs. B.SriRamya	Dr. K. Shirin Bhanu	Dr. K. Shirin Bhanu
5.	V18CST09	PP	Mr.Ch. Raja Ramesh	Mr. R. L. Phani Kumar	Mrs. A. Sirisha	Mr.R. L. Phani Kumar
6.	V18MBET5 1	MEFA	Dr.K.Rambabu	Mr. U.Bhargava	Dr.K.Rambabu	Mr. U.Bhargava
7.	V18CSL04	JP Lab	Mrs. B.SriRamya Mr B Madhava Rao Mrs D S L Manikanteswari	Mrs. B.SriRamya Mrs G Saraswathi Mrs. M Anantha Lakshmi	Dr. K. Shirin Bhanu Mrs. M Anantha Lakshmi Mrs D S L Manikanteswari	Dr. K. Shirin Bhanu Mr N V Murali Krishna raja Mrs B Sri Ramya
8.	V18CSL05	PP Lab	Mr. Ch. Raja Ramesh Dr. K. Shirin Bhanu Mrs G Loshma	Mr. R. L. Phani Kumar Mrs G Saraswathi Mr M Vamsi Krishna	Mrs. A. Sirisha Mr. R. L. Phani Kumar	Mr. R. L. Phani Kumar Mrs G Saraswathi Mrs. D. Sirisha
9.	V18CST61	TS-II	Mrs.A. Sirisha Mrs D S L Manikanteswari	Mr. V.K.HanumanTuraga Mr. K Venkatesh	Mr.V.K.HanumanTuraga Mr. K Venkatesh	Mrs. A. Sirisha Mr. K Venkatesh
10.	V18ENT11	COI	Dr.GCh S Madhusudhanarao	Dr.GCh S Madhusudhanarao	Dr.GCh S Madhusudhanarao	Mr. D. Satish
11.	V18ENT04	PCS-II	Mr. D. Naveen Kumar	Mr. D. Naveen Kumar	Mr. D. Naveen Kumar	Mr. D. Naveen Kumar



Head of the Department

COURSE STRUCTURE

IV-Semester

IV - Semester							
S.No	Course Code		Course	L	T	P	C
1	V18CST05	PCC	Computer Organization	3	0	0	3
2	V18CST06	PCC	Software Engineering	3	0	0	3
3	V18CST07	PCC	Formal Languages and Automata Theory	3	0	0	3
4	V18CST08	PCC	Java Programming	3	0	0	3
	V18CST09	PCC	Python Programming	3	0	0	3
5	V18MBET51	HSS	Managerial Economics and Financial Analysis	3	0	0	3
6	V18CSL04	PCC	Java Programming Lab	0	0	3	1.5
7	V18CSL05	PCC	Python Programming Lab	0	0	3	1.5
8	V18ENT11		Constitution of India	2	0	0	MNC
9	V18ENT04		Professional Communication Skills – II	3	0	0	MNC
10	V18CST61		Technical Skills -II	4	0	0	MNC
Total				27	0	6	21

Total Contact Hours: 33



LESSON PLANS

COMPUTER ORGANIZATION

Academic Year : 2019-20

Semester : IV

Name of the Course: COMPUTER ORGANIZATION

Programme: B.Tech

Sections :A,B,C&D

Course Code: V18CST05

COURSE OUTCOMES (Along with Knowledge Level):

After completion of this course, the students will be able to:

CO1: Illustrate Basic structure of Computers, Instruction types and their addressin mode[**K2**]

CO2: Describe the different modes of Input / Output transfer.[**K2**]

CO3: Illustrate different types of Memory. [**K2**]

CO4: Describe the different types of Control Unit techniques.[**K2**]

CO5: Illustrate the Fixed point and Floating point arithmetic operations of ALU. [**K2**]

CO6: Explain the concept of Pipelining.[**K2**]

Text Books:

1. Computer Organization, Carl Hamacher, Zvonko Vranesic, Safwat Zaky, 5th Edition, McGraw Hill Education.
2. Computer System Architecture, M. Morris Mano, 3rd Edition, Pearson Education.

Reference Books:

1. Computer Organization and Architecture, William Stallings, 10th Edition, Pearson Education.
2. Computer Architecture and Organization, John P. Hayes, 3rd Edition, McGraw Hill Education.

Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):

Course Outcome	Targeted Proficiency Level (% of Marks)	Targeted level of Attainment (% Students)
CO1	60	65
CO2	60	65
CO3	60	65
CO4	60	60
CO5	60	60
CO6	60	65

LESSON PLAN

CO1

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 1	Discuss CO's and syllabus of the course	K2	1	Lecture	BB
2		.Discuss the various functional units of computers	K2	2	Lecture	BB
3		Explain the Basic operational concepts	K2	1	Lecture with Discussion	BB
4		Discuss the Bus structure of computers	K2	1	Lecture with Discussion	BB
5		Explain the instructions and instruction Sequencing	K2	3	Lecture with Discussion	BB
6		Explain the various addressing modes of instructions	K2	2	Lecture with Discussion and in class Assignment	BB
7		Discuss the basic input/output operations of computers	K2	2	Lecture with Discussion	BB

12

CO2

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 2	Discuss the Accessing of I/O Devices	K2	1	Lecture with Discussion	BB
2		Explain the interrupt of I/O Devices	K2	2	Lecture with Discussion	BB
3		Discuss the Direct Memory Access	K2	1	Lecture with Discussion	BB
4		Explain the DMA Controller	K2	1	Lecture with Discussion	BB
5		Explain the DMA Transfer	K2	1	Lecture with Discussion	BB
6		Explain the Buses and its types	K2	3	Lecture with Discussion and in class Assignment	BB

09

CO3

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	Discuss the memory Hierarchy	K2	2	Lecture with Discussion	BB
2		Explain the main memory ram and roms chips	K2	1	Lecture with Discussion	BB
3		Discuss the memory address map	K2	1	Lecture with Discussion and in class Assignment	BB
4		Discuss the memory connection to CPU	K2	1	Lecture with Discussion	BB+ICT
5		Discuss the auxiliary memory used in computer	K2	2	Lecture with Discussion	BB+ICT
6		Discuss the Associative memory	K2	1	Lecture with Discussion	BB
7		Explain the hardware organization of associative memory	K2	1	Lecture with Discussion	BB+ICT
8		Explain the read and write operation of associative memory	K2	1	Lecture with Discussion	BB
9		Discuss the cache memory organization of Computer	K2	1	Lecture with Discussion	BB
10		Explain the mapping function of cache organization	K2	2	Lecture with Discussion	BB

13

CO4

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 4	Discuss the fundamental concepts in execution of instruction.	K2	1	Lecture	BB
2		Discuss the execution of instruction involves register transfer	K2	1	Lecture with Discussion	BB
3		Explain the process of performing an arithmetic or Logic operation	K2	1	Lecture with Discussion	BB
4		Discuss the process of fetching and storing a word from and to memory	K2	1	Lecture with Discussion	BB
5		Discuss the execution of a complete instruction	K2	2	Lecture with Discussion and in class Assignment	BB+ICT
6		Explain the multiple Bus organization	K2	1	Lecture with Discussion	BB
7		Discuss the Hardware control signals	K2	1	Lecture with Discussion	BB
8		Discuss the microprogrammed control Signals	K2	1	Lecture with Discussion	BB
9		Discuss the microinstructions in microprogrammed control unit	K2	1	Lecture with Discussion	BB
10		Discuss the microprogram sequencing	K2	1	Lecture with Discussion	BB

11

CO5

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 5	Discuss the representation of data for arithmetic instruction	K2	1	Lecture with Discussion	BB
2		Explain the Addition and subtraction of signed magnitude data with hardware implementation and hardware algorithm	K2	2	Lecture with Discussion	BB
3		Explain the Addition and subtraction of signed -2's complement data	K2	1	Lecture with Discussion	BB
4		Explain the multiplication of signed magnitude data with hardware implementation and hardware algorithm	K2	2	Lecture with Discussion	BB+ICT
5		Explain the booth multiplication algorithm	K2	1	Lecture with Discussion	BB
6		Explain the array multiplier	K2	1	Lecture with Discussion and in class Assignment	BB
7		Explain the Division of signed magnitude data with hardware implementation and hardware algorithm	K2	1	Lecture with Discussion	BB
8		Discuss the divide overflow	K2	1	Lecture with Discussion	BB
9		Discuss the floating-point arithmetic operations	K2	1	Lecture	BB
10		Explain the addition and subtraction of floating-point data	K2	1	Lecture with Discussion	BB
11		Explain the multiplication and division of floating-point data	K2	1	Lecture with Discussion	BB

13

CO6

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO6	Discuss the basic concepts of pipelining	K2	1	Lecture	BB
2		Discuss the role of cache memory	K2	1	Lecture	BB
3		Explain the pipe line performance	K2	2	Lecture with Discussion and in class Assignment	BB
4		Discuss the data hazards	K3	1	Lecture	BB
5		Discuss the operand forwarding in data hazards	K3	1	Lecture with Discussion	BB
6		Discuss the handling data hazards in software and its side effects	K3	1	Lecture with Discussion	BB
7		Discuss the instruction hazards	K2	1	Lecture	BB
8		Discuss the unconditional branches in instruction hazards	K2	1	Lecture	BB
9		Discuss the conditional branches and branch predication in instruction hazards	K2	1	Lecture	BB
10		Explain the datapath and control considerations in pipelined execution	K3	1	Lecture with Discussion	BB

11**Total No. of Classes: 69**

SOFTWARE ENGINEERING

Academic Year: 2019-20

Year/ Semester: II-II

Name of the Course: Software Engineering

After completion of this course, the student will be able to:

Programme: B.Tech

Section: A, B, C & D

Course Code: : V18CST06

CO	Course Outcomes	Knowledge Levels
V18CST06.1	Demonstrate Software Process Models.	K3
V18CST06.2	Illustrate Requirement Engineering Process.	K3
V18CST06.3	Discuss Software architecture and Design.	K2
V18CST06.4	Apply Coding principles and Testing techniques.	K3
V18CST06.5	Discuss Software Estimation and Maintenance.	K2
V18CST06.6	Describe Quality Management and Metrics.	K2

TEXT BOOKS:

- T1. Software Engineering, A practitioner's Approach- Roger S.Pressman, 7th Edition, McGrawHill International Edition
- T2. Software Engineering, 9/e, Sommerville, Pearson.
- T3. Software Engineering, A Precise approach, PankajJalote, Wiley

REFERENCE BOOKS:

- R1. CMMI and Six Sigma: Partners in Process Improvement , Jeannine M. Sivi, M. Lynn Penn, Robert W. Stoddard, 1st edition, Addison Wesley.
- R2. Software Engineering principles and practice, W S Jawadekar, 3rdEdition,TMH.

Targeted Proficiency Level :

Course Outcome	Targeted Proficiency Level (% of Marks)
1	65
2	65
3	65
4	60
5	60
6	60

Targeted Level of Attainment:

Course Outcome	Targeted level of Attainment (% Students)
1	65
2	65
3	65
4	60
5	60
6	60

Unit-I

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 1	Dissemination of Department Vision, Mission, PEOs, POs, PSOs, COs	-	1	Lecture	BB
2		Defining Software, Software application Domains	K1	1	Lecture	BB
3		Describe Legacy software. Software engineering,	K1	2	Lecture	BB
4		Describe the software process,	K1	1	Lecture with Discussion and in class Assignment	BB
5		Describe software Myths.	K1	1	Lecture	BB
6		Demonstrate Waterfall model, Prototyping	K3	2	Lecture	BB
7		Demonstrate Iterative development,	K3	1	Lecture	BB
8		Demonstrate Unified process	K3	1	Lecture	BB
9		Demonstrate Extreme programming and agile process	K3	1	Lecture with Discussion and in class Assignment	BB
		Explain Merits and Demerits of Software Process Models.	K2	1	Lecture	BB

Unit-II

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 2	Illustrate Functional and non-functional requirements	K3	2	Lecture with Discussion	BB
2		Illustrate User requirements, System requirements	K3	1	Lecture with Discussion	BB
3		Explain Interface specification	K2	1	Lecture with Discussion	BB
4		Explain the Software requirements document	K2	1	Lecture with Discussion and in class Assignment	BB
5		Explain Feasibility studies	K2	1	Lecture	BB
6		Explain Requirements elicitation and analysis	K2	1	Lecture	BB
7		Explain Requirements validation	K2	1	Lecture with Discussion and in class Assignment	BB
8		Explain Requirements management	K2	1	Lecture	BB

Unit-III

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	Explain Role of software architecture, Architecture views	K2	1	Lecture	BB
2		Describe components and connector view	K1	1	Lecture	BB
3		Explain architecture styles for C & C view	K2	2	Lecture with Discussion and in class Assignment	BB
4		Explain documenting architecture design	K2	1	Lecture	BB
5		Explain Evaluating architectures.	K2	1	Lecture	BB
6		Explain Design concepts	K2	2	Lecture with Discussion and in class Assignment	BB+ICT
7		Explain Function-oriented design	K2	1	Lecture	BB+ICT
8		Explain Object oriented design	K2	2	Lecture	BB+ICT
9		Describe Detailed design.	K1	1	Lecture	BB

Unit-IV

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 4	Demonstrate Programming principles and guidelines	K3	1	Lecture with Discussion	BB
2		Explain incrementally developing code, managing evolving code	K2	2	Lecture with Discussion	BB
3		Explain Testing concepts	K2	1	Lecture	BB
4		Explain testing process	K2	1	Lecture	BB
5		Demonstrate Black- box testing,	K3	1	Lecture with Discussion and in class Assignment	BB
6		Demonstrate White-box testing.	K3	1	Lecture	BB+ICT
7		Differentiate Reactive vs. Proactive Risk strategies	K2	1	Lecture	BB
8		Describe Software risks, Risk identification	K2	1	Lecture	BB
9		Explain Risk projection, Risk refinement	K2	1	Lecture	BB
10		Explain RMMM Plan.	K2	1	Lecture with Discussion and in class Assignment	BB

Unit-V

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 5	Explain Decomposition techniques	K2	1	Lecture	BB
2		Describe Empirical Estimation Models.	K1	2	Lecture with Discussion and in class Assignment	BB
3		Discuss Software Maintenance: Maintenance Process	K2	1	Lecture with Discussion and in class Assignment	BB
4		Explain Reverse Engineering	K2	1	Lecture	BB
5		Explain Reengineering	K2	1	Lecture	BB
6		Explain Configuration Management	K2	1	Lecture with Discussion	BB

Unit-VI

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 6	Describe Software Measurement and Metrics for software quality.	K2	1	Lecture with Discussion and in class Assignment	BB
2		Explain Quality concepts, Software quality assurance	K2	1	Lecture	BB
3		Distinguish Software Reviews and Formal technical reviews,	K2	2	Lecture with Discussion	BB
4		Explain Statistical Software quality Assurance and Software reliability	K2	2	Lecture	BB
5		Explain SEI-CMM Model,	K2	1	Lecture with Discussion and in class Assignment	BB+ICT
6		Explain Six Sigma	K2	1	Lecture	BB+ICT
7		Explain ISO 9000 quality standards.	K2	1	Lecture	BB+ICT

Total No. of Classes: 60

FORMAL LANGUAGES AND AUTOMATA THEORY

Academic Year: 2019-20

Programme: B.Tech

Year/ Semester: IV Semester

Section: A,B,C & D

Name of the Course: FLAT

Course Code: V18CST07

COURSE OUTCOMES (Along with Knowledge Level):

After completion of this course, the students will be able to:

- CO1. Construct DFA, NFA and ϵ -NFA. (K3).
- CO2. Produce Regular expressions, Regular Grammars. (K3)
- CO3. Construct Context Free Grammars, Context Free Languages. (K3).
- CO4. Construct Pushdown Automata and its equivalence with CFG. (K3).
- CO5. Construct Turing machine. (K3).
- CO6. Discuss Decidability Theory. (K2).

TEXT BOOKS:

1. Introduction to Automata Theory, Languages and Computation, J.E.Hopcroft, R. Motwani And J.D.Ullman, 3rd Edition, Pearson, 2008
2. Theory of Computer Science-Automata, Languages and Computation, K.L.P.Mishra and N.Chandrasekharan, 3rd Edition, PHI, 2007.

REFERENCE BOOKS:

1. Formal Language and Automata Theory, K.V.N.Sunitha and N.Kalyani, Pearson, 2015.
2. Introduction to Automata Theory, Formal Languages and Computation, ShyamalenduKandar, Pearson,2013.
3. Theory of Computation, V.Kulkarni, Oxford University Press, 2013.
4. Theory of Automata, Languages and Computation, Rajendra Kumar, McGraw Hill, 2014.

Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):

Course Outcome	Targeted Proficiency Level (% of Marks)	Targeted level of Attainment (% Students)
1	60	60
2	60	60
3	60	60
4	60	60
5	60	60
6	60	60

CO 1

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 1	Dissemination of Vision , Mission & PEO's Define Automata Theory?	K1	1	LECTURE	BB
2		Examine the Central Concepts of Automata Theory	K1	1	LECTURE	BB
3		State Automation, and Finite Automation	K1	1	LECTURE	BB
4		Describe about Transition Systems	K2	1	LECTURE	BB
5		Compute the acceptance of a String by a Finite Automation	K3	1	LECTURE	BB
6		Construct DFAs, DFA problems	K3	3	LECTURE	BB
7		Construct NFA,NFA problems	K3	1	LECTURE	BB
8		Illustrate the equivalence of DFA and NFA	K3	1	LECTURE	BB
9		Compute the conversion of NFA into DFA	K3	1	LECTURE	BB
10		Compute finite Automata with ϵ -Transition	K3	1	LECTURE	BB
11		Discover the minimization of Finite Automata	K3	1	LECTURE	BB
12		Explain about Mealy and Moore Machines	K2	2	Discussion	ICT
13		Describe the Applications and Limitation of Finite Automata	K1	1	LECTURE	BB
		TOTAL		16		

CO 2

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 2	Describe Regular Expressions	K1	1	LECTURE	BB
2		Outline Regular Sets, Identity Rules	K1	1	LECTURE	BB
3		Illustrate the Equivalence of two Regular Expressions	K3	1	LECTURE	BB
4		Describe Finite Automata, and Regular Expressions	K2	1	Discussion	ICT
5		Illustrate the Equivalence between Finite Automata and Regular Expressions	K3	2	Discussion	ICT+BB
6		Describe Pumping Lemma, Closers/ Properties	K2	1	LECTURE	BB
7		Describe the Applications of Regular Expressions and , Finite Automata and Regular Grammars	K2	1	LECTURE	BB
8		Describe the Regular Expressions and Regular Grammars	K2	1	LECTURE	BB
		TOTAL		9		

CO 3

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	Define Formal Languages, Grammars	K1	1	LECTURE	BB
2		Classify the Grammars, Chomsky Hierarchy Theorem	K2	1	LECTURE	BB
3		Describe Context Free Grammar, Examples	K2	1	LECTURE	BB
4		Illustrate the Leftmost and the Rightmost Derivations, Parse Trees with examples	K3	1	Discussion with Lecture	ICT
5		Explain about the Ambiguous Grammars	K2	1	LECTURE	BB
6		Determine the Simplification of Context Free Grammars- Elimination of Useless Symbols, E-Productions and Unit Productions with examples	K3	2	Discussion	BB
7		Illustrate the Normal Forms for Context Free Grammars- Chomsky Normal Form and Greibach Normal Form	K3	2	ICT	BB
8		Explain about Pumping Lemma, Closure Properties	K2	1	LECTURE	BB+ICT
9		Describe the Applications of Context Free Grammars	K1	1	LECTURE	BB
		TOTAL		11		

CO 4

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 4	Define Pushdown Automata and, Model	K1	1	LECTURE	BB
2		Describe the Graphical Notation, Instantaneous Description Language Acceptance of pushdown Automata	K2	1	LECTURE	BB
3		Constructthe Pushdown Automata	K3	2	Lecture with Discussion	BB+ICT
4		Illustrate deterministic and Non – Deterministic Pushdown Automata	K2	1	LECTURE	BB
5		Compute the Equivalence of Pushdown Automata and Context Free Grammars Conversion	K3	1	Discussion	BB
6		Construct Two Stack Pushdown Automata, Describe the Applications of Pushdown Automata	K3	2	LECTURE	BB+ICT
		TOTAL		8		

CO 5

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 5	Define Turing Machine	K1	1	LECTURE	BB
2		Define the Model, Representation of Turing Machines-Instantaneous Descriptions	K1	1	LECTURE	BB
3		Explain the Transition Tables and Transition Diagrams with examples	K2	2	LECTURE	BB
4		Explain the Language of a Turing Machine	K2	1	LECTURE	BB
5		Estimate the Language of a Turing Machine examples	K3	1	LECTURE	BB
6		Construct a Turing Machines and, Techniques for Turing Machine	K3	2	LECTURE	BB

7		Discuss the Types of Turing Machines	K2	1	LECTURE	BB
8		Describe the Church's Thesis, Universal Turing Machine, Explain the Restricted Turing Machine	K2	1	LECTURE	BB
		TOTAL		10		

CO 6

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 6	Describe about the Decidable and Undecidable Problems	K1	1	LECTURE	BB
2		Describe the Halting Problem of Turing Machines	K2	1	LECTURE	BB
3		Review the Post's Correspondence Problem	K2	1	LECTURE	BB
4		Describe the Modified Post's Correspondence Problem	K2	1	LECTURE	BB
5		Classify the Classes of P and NP	K2	1	LECTURE	BB
6		Classify the NP-Hard and NP-Complete Problems	K2	1	LECTURE	BB
		TOTAL		6		

Total No. of Classes: 60

JAVA PROGRAMMING

Academic Year: 2019-20

Programme: B.Tech

Year/ Semester: IV Semester

Section: A,B,C & D

Name of the Course: Java Programming

Course Code: V18CST08

COURSE OUTCOMES (Along with Knowledge Level):

After successful completion of the Course, the student will be able to:

CO1: Describe Java Virtual Machine and Type casting. [K2]

CO2: Demonstrate Concepts like Constructors, Arrays, Nested Classes and Command Line Arguments. [K3]

CO3: Implement concepts of Inheritance and Exception Handling. [K3]

CO4: Develop programs on Multi-Threading and Files. [K3]

CO5: Demonstrate Applet Programming and AWT Components. [K3]

CO6: Demonstrate Event Handling and Swings. [K3]

Text Books:

1. Java Programming, E.Balagurusamy, 4th Edition, TMH.
2. The complete Reference Java, 8th Edition, Herbert Schildt, TMH.
3. Introduction to java programming, Y Daniel Liang, 7th Edition, Pearson.

Reference books:

1. Core Java: An Integrated Approach, R Nageswara Rao, 7th Edition, DreamTech
2. Head First Java, Kathy Sierra and Bert Bates, 2nd Edition O'reilly

Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):

Course Outcome	Targeted Proficiency Level (% of Marks)	Targeted level of Attainment (% Students)
CO1	65	65
CO2	65	65
CO3	65	65
CO4	60	60
CO5	60	60
CO6	60	60

LESSON PLAN

CO 1

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO1	Recall the Need of Object Oriented Programming and the Principles of Object Oriented Languages	K1	1	Lecture+ Discussion	BB
3		Explain different applications of OOP	K2	1	Lecture+ Discussion	BB
4		Describe the history of Java	K2	1	Lecture	BB
5		Discuss about different features of java	K2	1	Lecture	BB
6		Explain about Java Virtual Machine	K2	1	Lecture	BB
7		Discuss about Java Program Structure	K2	1	Lecture	BB
8		Discuss about Variables, Primitive Datatypes and Identifiers	K2	1	Lecture	BB
9		Discuss about constants, Operators and Expressions	K2	2	Lecture	BB
10		Discuss Precedence Rules and Associativity	K2	1	Lecture	BB
11		Illustrate Primitive Type conversion and casting with an example	K2	1	Lecture	BB
12		Discuss briefly about control structures	K2	1	Lecture	BB

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CO 2

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO2	Explain about classes and objects	K2	2	Lecture+ Discussion	BB
2		Discuss about methods	K2	1	Lecture	BB
3		Describe Constructors and Constructor Overloading	K2	2	Lecture+ Discussion	BB
4		Illustrate the use of this keyword with examples	K2	1	Lecture	ICT
5		Discuss about the importance of Static keyword	K2	1	Lecture	BB
6		Explain about different types of Arrays	K2	2	Lecture	BB
7		Interpret Command line arguments	K3	1	Lecture+ Discussion	ICT
8		Demonstrate Nested Classes	K3	1	Lecture	BB

CO 3

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO3	Explain about different types of Inheritance	K2	1	Lecture	BB
2		Illustrate the use of super keyword and final keyword	K2	1	Lecture	BB
3		Explain the concept of Method Overriding	K2	1	Lecture	BB
4		Explain about Abstract class	K2	1	Lecture	BB
5		Explain about interface	K2	1	Lecture	BB
6		Illustrate the procedure of creating packages and using packages	K2	1	Lecture	ICT
7		Explain the importance of CLASSPATH	K2	1	Lecture	ICT
8		Describe different types of Exceptions and procedure of Exception Handling	K2	3	Lecture	BB
9		Construct programs using Exception handling techniques like try...catch and finally block	K3	3	Lecture + Discussion	BB
10		Interpret throw and throws statements with examples	K3	1	Lecture	BB

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CO 4

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO4	Discuss about thread lifecycle	K3	1	Lecture	BB
2		Interpret Thread Priorities and Thread Synchronization with examples	K3	2	Lecture	ICT
3		Illustrate Communication between threads with example program	K3	1	Lecture	ICT
4		Illustrate various file operations like Reading data from and writing data to files	K3	3	Lecture	BB
5		Demonstrate Random Access Files	K3	1	Lecture	BB

8

CO 5

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 5	Explain about applet class	K2	1	Lecture	BB
2		Discuss about Applet Lifecycle	K2	1	Lecture	BB
		Discuss about AWT ,Components and Containers of AWT	K2	1	Lecture+ Discussion	ICT
		Illustrate various AWT Controls like Button,label,Checkbox, RadioButton,List box, Menu and Scrollbar with example programs	K3	4	Lecture+ Discussion	ICT
		Interpret different types of layout managers with examples	K3	2	Lecture+ Discussion	ICT

9**CO 6**

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 6	Describe Event Delegation Model, Source of Events and Event Listeners	K3	1	Lecture	ICT
2		Illustrate Adapter classes and inner classes with example programs	K3	2	Lecture	ICT
3		Distinguish AWT and swings	K3	1	Lecture + Discussion	ICT

4**Total No. of hours: 58**

PYTHON PROGRAMMING

Academic Year: 2019-20

Year/ Semester: II-II

Name of the Course: Python Programming

Course Outcomes (Along with Knowledge Level):

After Completing the course Student will be able to:

Programme: B.Tech

Section: A, B, C & D

Course Code: : V18CST09

S.No	Course Outcomes	Knowledge
CO1	Illustrate the basic concepts of python programming	K2
CO2	Describe the control structures of python.	K2
CO3	Demonstrate functions and packages.	K3
CO4	Construct python programs using structured data types	K3
CO5	Compare Text Files and Binary Files.	K4
CO6	Apply OOPs concepts to Develop Test cases.	K3

TEXT BOOKS

1. “Python Programming using problem solving Approach” Reema Thareja, Oxford University Press – 2017.

2. Python with Machine Learning by A.Krishna Mohan, Karunakar & T.Murali Mohan by S. Chand Publisher-2018. **Reference Books:**

1. Allen B. Downey, “Think Python: How to Think Like a Computer Scientist“, 2nd edition, Updated for Python 3, Shroff /O’Reilly Publishers, 2016 .

2. Guido van Rossum and Fred L. Drake Jr, An Introduction to Python – Revised and updated for Python 3.2, Network Theory Ltd., 2011.

3. John V Guttag, “Introduction to Computation and Programming Using Python“, Revised and expanded Edition, MIT Press , 2013

4. Robert Sedgewick, Kevin Wayne, Robert Dondero, “Introduction to Programming in Python:

Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):

Course Outcome	Targeted Proficiency Level (% of Marks)	Targeted level of Attainment (% Students)
CO1	60	60
CO2	60	60
CO3	60	60
CO4	60	60
CO5	60	60
CO6	60	60

CO1: UNIT-I

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO1	Discuss Blooms Taxonomy	K1 and K2	1	LECTURE	BB
2		Explain History of Python	K2	1	LECTURE	BB
3		Discuss Features of python	K2	1	LECTURE	BB
4		Describe Future of Python installation and execution Show Running Python Scripts	K2	1	LECTURE	BB
5		Explain Data types	K2	1	LECTURE	BB
6		Explain Variables, Assignment, Keywords, Input-Output, Indentation.	K2	1	LECTURE with Discussion	BB
7		Explain type conversions	K2	1	LECTURE with Discussion	BB
8		Discuss Literal constants, Numbers, Strings	K2	1	LECTURE	BB
9		Explain Operators and expressions, operator precedence and expression evaluation	K2	1	LECTURE	BB
		TOTAL		9		

CO2: UNIT-II

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO2	Discuss with decision control statements if alternative if-else	K2	1	LECTURE	BB
2		Discuss chained conditional (if-elif-else):	K2	2	Discussion	ICT+BB
3		Discuss with Control Structures: While loop	K2	1	Discussion	ICT+BB
4		Explain for loop	K2	1	Discussion	ICT+BB
5		Explain nested for loop	K2	2	Discussion	ICT+BB
6		Discuss with range function	K2	1	Discussion with Lecture	BB
7		Discuss with break	K2	3	Discussion with Lecture	ICT+BB

8		Discuss with continue	K2	1	Discussion with Lecture	BB
9		Discuss with pass statements	K2	1	Discussion with Lecture	ICT+BB
		TOTAL		13		

CO3: UNIT-III

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO3	Explain Functions	K2	1	LECTURE	BB
2		Discuss with Function declaration and Definition	K2	1	LECTURE	BB
3		Explain Function Call	K2	2	LECTURE	BB
4		Discuss with variable Scope and Lifetime	K2	1	Lecture	ICT
5		Explain the return statement	K2	1	LECTURE	BB
6		Explain Lambda functions	K2	1	Discussion	BB
7		Explain Anonymous functions	K2	1	ICT	BB
8		Discuss Documentation strings	K2	1	Lecture	ICT
9		Demonstrate with Modules	K3	2	Lecture	BB
10		Demonstrate with Packages	K2	1	Lecture	ICT
11		Practice Exercises	K2	1	Lecture	BB
		TOTAL		13		

CO4: UNIT-IV

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO4	Defining Lists	K2	1	LECTURE	BB
2		Discuss List operations	K2	1	Lecture with Discussion	BB
3		Discuss List slices	K2	1	Lecture with Discussion	BB+ICT
4		Explain List methods	K2	1	LECTURE	BB
5		Demonstrate cloning lists	K3	1	Discussion	BB
6		Discuss list parameters	K2	2	Discussion	BB
7		Discuss with Tuples	K2	1	LECTURE	BB
8		Explain tuple assignment and tuple as return value	K2	1	LECTURE	BB
9		Discuss with set creation and set operations	K2	1	LECTURE	BB
10		Explain Dictionaries creation and operations	K2	1	LECTURE	BB
11		Create comprehension, operations on strings	K3	2	LECTURE	BB
		TOTAL		13		

CO5: UNIT-V

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO5	Define File	K1	1	LECTURE	BB
2		Types of files	K1	1	LECTURE	BB
3		Explain File handling	K2	2	LECTURE	BB
4		Explain Text Files reading and writing	K2	1	LECTURE	BB
5		Discuss Error and Exceptions introduction	K2	1	Discussion	BB+ICT
6		Discover Difference between an error and Exception, Handling Exception	K3	1	LECTURE	BB
7		Explain try except block, Raising Exceptions	K2	2	LECTURE	BB
		TOTAL		9		

CO6: UNIT-VI

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO6	Define OOPs concepts	K1	1	LECTURE	BB+ICT
2		Explain classes	K2	1	LECTURE	BB
3		Explain Methods	K2	2	LECTURE	BB
4		Demonstrate Constructor	K3	2	LECTURE	BB+ICT
5		Discuss with inheritance	K2	2	LECTURE	BB+ICT
6		Discuss Overriding methods	K2	1	LECTURE	BB
7		Discuss Data hiding	K2	1	LECTURE with Discussion	BB
8		Explain GUI programming with Tkinter	K2	1	LECTURE	BB+ICT
		Total		11		

Total Number of Hours= 68

Managerial Economics and Financial Analysis

Academic Year: 2019-20

Programme: B.Tech

Semester : 4th Semester

Section: A, B, C&D

Name of the Course: MEFA

Course Code: V18MBT51

Name of the Course Instructors: Dr. K.Rambabu & Sri U. Bhargava

Course Outcomes (Along with Knowledge Level):

CO	K_x	Course Outcome
V18MBT51.1	K ₂	Equipped with the knowledge of estimating the demand for a product and the relationship between price and demand.
V18MBT51.2	K ₃	Understands the role of production concept, least cost combinations and various costs concepts in decision making
V18MBT51.3	K ₂	Understands the price –output determination under various market structures
V18MBT51.4	K ₂	Equipped with the knowledge of various forms of business organization
V18MBT51.5	K ₃	One will be able to prepare the financial statements and also gains knowledge to analyze them.
V18MBT51.6	K ₃	Equipped with knowledge on taking capital budgeting decisions and knows how to use those capital budgeting methods

Text Books/ Reference Books suggested:

1. Aryasri: Managerial Economics and Financial Analysis, 2/e, TMH, 2005. 2.
2. Varshney&Maheswari: Managerial Economics, Sultan Chand, 2003.
3. S A Siddhiqui&AS Siddhiqui , Managerial Economics and Financial Analysis, New Age international publishers 2013.

Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):

Course Outcome	Targeted Proficiency Level (% of Marks)	Targeted level of Attainment (% Students)
1	60	60
2	60	60
3	60	60
4	60	60
5	60	60
6	60	60

S.no	Course Outcome	Contents	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids	Remarks
	Estimate the demand for a product and the relationship between price and demand.	UNIT-Introduction to managerial economics		10			
1		Define managerial economics	K1	1	Lecture Discuss	Blackboard	
2		Describe ME with other disciplines	K1	1	Lecture	Blackboard	
3		Explain Nature and scope of managerial economics	K2	1	Lecture	Blackboard	
4		Define Demand	K1	1	Lecture Discuss	Blackboard	
5		Describe law of demand	K2	1	Lecture	Blackboard	
6		Explain Elasticity of demand	K2	1	Lecture	Blackboard	
7		Find the of elasticity of demand	K2	2	Lecture	Blackboard	
8		Explain Demand forecasting, methods.	K2	2	Lecture Discuss	Blackboard	
		UNIT-II: Production analysis and cost analysis		08			
1	Interpret production concept, least cost combinations and various costs concepts in decision making	State Production function	K1	1	Lecture	Blackboard	
2		State Isocost	K1	1	Lecture	Blackboard	
3		State Iso quants	K1	1	Lecture	Blackboard	
4		Explain Cob-Douglas production function	K2	1	Lecture Discuss	Blackboard	
5		Describe economies of scale	K2	1	Lecture	Blackboard	
		Enumerate various cost concepts	K1	1	Lecture	Blackboard	
		Solve break even analysis	K3	2	Lecture	Blackboard	
		UNIT-III: market structures		10			

1	Describe the price - output determination under various market	Describe Different types of market structures	K1	3	Lecture	Blackboard	
2		Explain Price-output determination	K2	3	Lecture	Blackboard	
3		Explain Pricing methods	K2	2	Lecture	Blackboard	
4		Describe Theory of firms	K2	2	Lecture	Blackboard	
		UNIT-IV: Business organization		06			
1	Classify various forms of	State Features, merits ,demerits of Different forms of Organizations	K1	4	Lecture	Blackboard	
2		Differentiate various forms of business organizations	K2	2	Lecture	Blackboard	
		Unit-V: financial accounting		12			
1	Prepare the financial statements to gains knowledge to analyze	Describe Accounting Terminology	K1	1	Lecture	Blackboard	
		Explain Double Entry System	K2	1	Lecture	Blackboard	
		Prepare of financial statements	K3	4	Lecture	Blackboard	
2		Examine of financial statements by using funds flow, Ration ,	K3	6	Lecture	Blackboard	
		UNIT-VI: capital budgeting		08			
1	Assess various investment project proposals with the	Define Capital	K1	1	Lecture	Blackboard	
2		Enumerate types of capital	K1	1	Lecture	Blackboard	
3		Explain capital budgeting, Process	K2	2	Lecture	Blackboard	
5		Apply capital budgeting techniques	K3	4	Lecture	Blackboard	

Total No. of Classes: 54

Java Programming Lab

Academic Year : 2019-20

Programme:B.Tech

Semester:IV

Branch:CSE

Sections:A,B,C& D

Name of the Course:Java Programming Lab

Course Code: V18CSL04

COURSE OUTCOMES:

Course Outcomes (Along with Knowledge Level):

After completing this course the student able to:

V18CSL04.1	Develop Programs to handle on Classes and Objects.	[K3]
V18CSL04.2	Demonstrate Constructors and Arrays.	[K3]
V18CSL04.3	Demonstrate Inheritance and Exception Handling.	[K3]
V18CSL04.4	Implement programs on Multi-Threading.	[K3]
V18CSL04.5	Illustrate File Handling Mechanisms.	[K3]
V18CSL04.6	Demonstrate GUI Programming using Applets and Swings.	[K3]

Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):

Course Outcome	Targeted Proficiency Level (% of Marks)	Targeted level of Attainment (% Students)
CO1	60	65
CO2	60	65
CO3	60	65
CO4	60	60
CO5	60	60
CO6	60	65

LESSON PLAN

CO 1

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO1	Lab 1: Develop java programs on Control Structures and Type Conversions in java	K3	6	Lecture & Experiment	ICT
2		Lab 2: Develop java programs using the following concepts a. classes and objects b. static keyword				

CO2

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
3	CO2	Lab 2: Develop java programs to implement constructors	K3	6	Lecture & Experiment	ICT
4		Lab 3: Develop java programs using the following concepts. a) Arrays b) Nested Classes c) Command Line Arguments				

CO 3

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
5	CO3	Lab 4: Develop java programs using the following concepts. a) Inheritance b) Usage of super c) Method Overriding	K3	9	Lecture & Experiment	ICT
6		Lab 5: Develop java programs to implement various concepts like a) Usage of final keyword b) Abstract class c) Interfaces				
7		Lab 6: Experiment java programs on a) Packages b) Exception Handling.				

CO 4

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
8	CO4	Lab 7:Develop java programs to implement multi threading using various concepts like a.Multiple threads acting on single object b.Deadlock c.Thread Communication	K3	3	Lecture, Experiment	BB&ICT

CO 5

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
9	CO5	Lab 8: Develop java programs to perform reading and writing operations on sequential files and random access files	K3	6	Lecture, Experiment	ICT

CO 6

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
10	CO6	Lab 9:Develop java Programs to design GUI using Applets & AWT Components.	K3	9	Lecture, Experiment	BB&ICT
11		Lab 10:Develop java Programs toEvent Handling using Listener Interfaces.				
12		Lab 11:Develop java Programs to design GUI using Swings.				

Total no of hours:39

Python Programming Lab

Academic Year : 2019-20

Semester: II/IV

Branch: CSE

Name of the Course: Python Programming Lab

Programme:B.Tech

Sections:A,B,C& D

Course Code: V18CSL04

COURSE OUTCOMES:

Course Outcomes (Along with Knowledge Level):

After completing this course the student able to:

V18CSL04.1	Develop Programs to handle on Classes and Objects.	[K3]
V18CSL04.2	Demonstrate Constructors and Arrays.	[K3]
V18CSL04.3	Demonstrate Inheritance and Exception Handling.	[K3]
V18CSL04.4	Implement programs on Multi-Threading.	[K3]
V18CSL04.5	Illustrate File Handling Mechanisms.	[K3]
V18CSL04.6	Demonstrate GUI Programming using Applets and Swings.	[K3]

Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):

Course Outcome	Targeted Proficiency Level (% of Marks)	Targeted level of Attainment (% Students)
CO1	60	65
CO2	60	65
CO3	60	65
CO4	60	60
CO5	60	60
CO6	60	65

LESSON PLAN

CO 1

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO1	Exercise 1 - Basics a) A sample Python Script using command prompt, Python Command Line and IDLE b) A program to purposefully raise an Indentation Error and correct it Exercise 2 - Operations a) A program to compute distance between two points taking input from the user(Pythagorean Theorem) b) A program on add.py that takes 2 numbers as command line arguments and prints its sum.	K3	6	Lecture & Experiment	ICT

CO2

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
2	CO2	Exercise - 3 Control Flow a) A Program to implement for checking whether the given number is a even number or not. b) A program to construct reverses the digits of a given number and add it to the original, if the sum is not a palindrome repeat this procedure. c) A program using a while loop that asks the user for a number, and prints a countdown from that number to zero. Exercise 4 - Control Flow – Continued a) A program to construct the following pattern, using a nested for loop. <pre> *</pre> b) By considering the terms in the Fibonacci sequence whose values do not exceed four million, find the sum of the even-valued terms.	K3	3	Lecture & Experiment	ICT

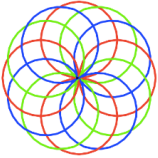
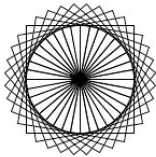
S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
3	CO3	Exercise - 5 – Problem Solving using Functions a) Find mean, median, mode for the given set of numbers passed as arguments to a function. b) Develop a function nearlyequal to test whether two strings are nearly equal. Two strings a and b are nearly equal when a can be generated by a single mutation on b c) Develop a Recursive Function to find the Factorial of a given number. d) Develop function to compute gcd, lcm of two numbers. Each function shouldn't exceed one line. Exercise - 8– Modules a) Install packages requests, flask and explore them using (pip) b) A program to implement a script that imports requests and fetch content from the page. Eg. (Wiki) c) Develop a simple script that serves a simple HTTPResponse and a simple HTML Page	K3	9	Lecture & Experiment	ICT

CO 4

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
4	CO4	Exercise - 6 Structured Data types a) a program to count the number of strings where the string length is 2 or more and the first and last character are same from a given list of strings. b) a program to develop unzip a list of tuples into individual lists and convert them into dictionary. Exercise – 7 Structured Data types Continued a) A program to count the numbers of characters in the string and store them in a dictionary data structure b)a program to use split and join methods in the string and trace a birthday with a dictionary data structure.	K3	6	Lecture, Experiment	BB & ICT

CO 5

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
5	CO5	Exercise - 9 Files a) a program to count frequency of characters in a given file. Can you use character frequency to tell whether the given file is a Python program file, C program file or a text file? b) a program to compute the number of characters, words and lines in a file.	K3	9	Lecture, Experiment	ICT

		Exercise - 10 OOP a) Class variables and instance variable and illustration of self variable i) Robot ii) ATM Machine Exercise - 11 GUI, Graphics 1. Develop a GUI for an Expression 2. A program to implement the following figures using turtle  				
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CO 6

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
6	CO6	Exercise - 12 - Testing a) Develop a test-case to check the even numbers b) Develop a test-case to check the function reverse string which returns reversed string Case Study: Implement Rock – paper – Scissors game using tkinter.	K3	3	Lecture, Experiment	BB&ICT

		Addon programs	K3	3	Lecture, Experiment	BB&ICT
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Total no of hours:39

BB* Black Board

ICT* Information Communication Technology

CONSTITUTION OF INDIA

Academic Year : 2019-20

Programme: B.Tech

Semester : IV

Sections :A,B,C&D

Name of the Course: CONSTITUTION OF INDIA

Course Code: V18ENT11

COURSE OUTCOMES (Along with Knowledge Level):

After completion of this course, the students will be able to:

CO1: Summarize the evolution and historical importance of the Indian Constitution from 1858 to 1947. **[K2]**

CO2: Explain various stages in the composition of the Indian Constitution. **[K2]**

CO3: Develop awareness about their primary rights and duties & build up their civic sense. **[K3]**

CO4: Explain the distribution of powers between the center and states. **(K2)**

CO5: Summarize and sketch the specific roles of heads of Nation and the functioning of legislative bodies. **(K2)**

CO6: Explain the role of local self-government in strengthening democracy. **(K1)**

Reference Books:

1. D D Basu-Introduction to the Constitution of India – 18th Edition. P.H.I.-New Delhi-1998
2. Granville Austin (1972) the Indian Constitution, Cornerstone of a Nation, Oxford university Press, New Delhi
3. Madhavkhosla (2012) the Indian Constitution, Oxford University Press, New Delhi
4. Granville Austin (1999) Working a Democratic Constitution; A History of the Indian Experience, Oxford University Press, New Delhi
5. Zoya Hasan, Sridharan E and Sudharshan R (Eds) 2002 India's living Constitution, Permanent black, New Delhi
6. Baxi Upendra (1980) the Indian Supreme Court and Politics, Eastern Book Co, Lucknow.

Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):

Course Outcome	Targeted Proficiency Level (% of Marks)	Targeted level of Attainment (% Students)
CO1	60	65
CO2	60	60
CO3	60	60
CO4	60	60
CO5	60	60
CO6	60	65

LESSON PLAN

CO1

Sl. No.	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 1	Discuss CO's and syllabus of the course & Historical Perspective of the Indian Constitution	K2	1	Lecture	BB
2		Explain Regulating Act of 1773, Pitt's India Act of 1784, & Charter Acts of 1833, 1853	K2	1	Lecture	BB
3		Explain Government of India Act of 1858, & Indian Councils Act of 1861, 1892, 1909	K2	1	Lecture with Discussion	BB
4		Explain Government of India Act of 1919, Simon Commission, & Communal Award	K2	1	Lecture	BB + ICT
5		Discuss the Government of India Act of 1935	K2	1	Lecture with Discussion	BB + ICT
6		Explain the Indian Independence Act of 1947	K2	1	Lecture with Discussion	BB

06

CO2

Sl. No.	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 2	Explain the Preparation of the Indian Constitution by Constituent Assembly of India	K2	1	Lecture	BB
2		Discuss the Preamble or Philosophy of the Indian Constitution	K2	2	Lecture with Discussion	BB
3		Discuss the Salient Features of the Indian Constitution	K2	2	Lecture with Discussion	BB + ICT

05

CO3

Sl. No.	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	Explain the Fundamental Rights - Their Importance and Limitations	K3	2	Lecture	BB + ICT
2		Discuss the Fundamental Duties and their Importance	K3	1	Lecture with Discussion	BB
3		Discuss the Directive Principles of the State Policy and their Implementation	K3	2	Lecture	BB + ICT

05

CO4

Sl. No.	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 4	Explain about Indian Federalism	K2	1	Lecture	BB
2		Discuss the Distribution of powers between Union and State Governments	K2	2	Lecture with Discussion	BB + ICT
3		Explain Legislative, Executive and Financial relations between Union and State Governments	K2	2	Lecture with Discussion	BB + ICT

05

CO5

Sl. No.	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 5	Explain the Parliamentary form of Government in India	K2	1	Lecture	BB
2		Discuss the Union Executive - a) President of India- Powers and functions	K2	1	Lecture with Discussion	BB + ICT
3		Discuss the Union Executive - b) Vice-President - Powers and functions, C) Prime Minister and Council of Minister - Powers and functions	K2	1	Lecture with Discussion	BB + ICT

4		Explain the Union Legislature - a) Rajya Sabha – Powers and Functions, b) Lok Sabha- Powers and Functions	K2	1	Lecture	BB + ICT
5		Explain Union Legislature - c) Amending Procedure- Important Constitutional Amendments – 42 nd , 44 Constitutional Amendment Acts.	K2	1	Lecture with Discussion	BB + ICT
6		Explain the Judiciary – Supreme court of India - Powers and Functions	K2	1	Lecture with Discussion	BB + ICT

06

CO6

Sl. No.	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 6	Explain the Local Self-government in India	K1	1	Lecture	BB + ICT
2		Explain the Local Self-government in India 73 rd Constitutional Amendment Act - Rural Areas Panchayati Raj System	K1	2	Lecture with Discussion	BB + ICT
3		Explain the Local Self-government in India 74 th Constitutional Amendment Act - Urban Areas Municipalities	K1	2	Lecture with Discussion	BB + ICT

05

Total No. of Classes: 32

Professional Communication Skills - II

S.No	Course Code	Course Name	L	T	P	C
1	V18ENT04	Professional Communication Skills - II	3	-	-	MNC

The students will be able to

CO1: Correlate individual words into one whole sentence using new vocabulary and focus on the error analysis of prepositions and conjunctions. [K4]

CO2: Distinguish and acquire knowledge of using words of same category in a sentence and learn new words that promote communicative finesse. [K5]

CO3: Find errors in sentences where the modifiers are misplaced and put them at the appropriate place, use hit pair words and send an email that is concise and lucid[K5]

CO 4: Interpret the importance of Attire and Etiquette in societal context and manage their time. (K2)

CO5: Discover the team working abilities among themselves and display their leadership qualities. (K3)

CO6: Identify various elements of emotional balance that have positive impact on work-life-balance. (K2)

Reference:

1. Essential English Grammar - Raymond Murphy
2. Advanced English Grammar - D.S. Paul
3. Word Power Made Easy - Norman Lewis
4. English collocations in use - Michael McCarthy
5. Word Power Made Handy - ShaliniVarma
6. Barron's GRE - Barron's
7. Current English Grammar & Usage – R.P Sinha
8. Think & Grow Rich - Napoleaon Hill
9. Soft Skills for Everyone - Butterfield, Jeff,
10. Soft Skills - Chauhan, G.S. and Sangeeta Sharma
11. Theories of Personality - Hall, Calvin S

12. Corporate Conversations - Holtz, Shel
13. . Communication Skills - Kumar, Sanajy and PushpLata
14. Winning at Interviews - Thorpe, Edgar and Showick Thorpe
15. Swami Vivekananda and "Personality Development" published by RK Math.

Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):

Course Outcome	Targeted Proficiency Level (% of Marks)	Targeted level of Attainment (% Students)
1	50	60
2	50	60
3	50	60
4	50	60
5	50	60
6	50	60

CO 1

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 1	Dissemination of CO s and Introduction of the course to the students along with model papers.		1	Lecture	BB/Handout
2		To identify and recognize the words using roots	K1	2	Discussion	A.V
3		Can analyze sentence errors of prepositions and conjunctions	K4	3	Lecture	BB/ Handout
5		Make use of foreign expressions in day to day communication.	K3	1	Lecture	BB/ Handout.

CO 2

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 2	Can identify mistakes in the usage of sentence structures and learn parallel constructions	K4	2	Lecture	BB/Handout
2		Can use synonymous words to refine their communication	K3	1	Team work	blackboard
3		Perceive the intended meaning of proverbs	K5	2	individual	BB/ handout
4		Can use synonyms and antonyms with ease	K3	1	Lecture	BB/ Hand out

CO 3

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	Can identify dangling modifiers in a sentence and rectify them	K3	2	Lecture	BB/workbook
2		Use Foreign expressions in conversation	K3	2	Pair work	workbook
3		Can write emails in a professional way	K5	1	individual	workbook
4		Take part in activities using the knowledge bank of idioms and phrases.	K3	1	Lecture	BB/ workbook.

CO4

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 4	Able to understand the importance of Attire and Etiquette	K2	2	Lecture	PPT
2		Can understand various principles and theories of managing Time and plan the activities effectively.	K3	2	Lecture	PPT

CO5

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 5	Take part in various Team building activities	K3	2	Role Play	PPT
2		Able to understand Team dynamics and how to build teams	K3	2	Activity	PPT
3		Can understand various leadership styles	K3	1	Lecture	BB
4		Will be able to analyze various personality traits and myths about leadership	K3	2	Lecture	BB.

CO6

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 6	Recognize the importance of balancing work and life	K2	2	Lecture	PPT
2		Understand various models of emotional intelligence	K2	2	Lecture	PPT

Total No. of Classes: 34

TECHNICAL SKILLS-II

Academic Year: 2019-20

Semester: IV

Name of the Course: TECHNICAL SKILLS-II

Programme: B.Tech

Section: A,B,C & D

Course Code:V18CST61

COURSE OUTCOMES (Along with Knowledge Level):

After completion of this course, the students will be able to:

CO1 :Develop programs using Pointers.(K3)

CO2 :Develop problems using functions.(K3)

CO3 : Solve problems using recursions.(K3)

CO4 :Constructprograms using File Handling.(K3).

CO5 :Develop programs using Structures and Unions (K3).

CO6 :Make use of command line arguments and preprocessors to solve the given problems(K3)

TEXT BOOKS:

1. Let us C: YesvanthKanetkar, BPB Publications, 16th Edition
2. Working With C, Yashavant P. Kanetkar, BPB Publications
3. Test Your C Skills, Yashavant P. Kanetkar, BPB Publications

Online Resources: [geeksforgeeks](https://www.geeksforgeeks.org/)

Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):

Course Outcome	Targeted Proficiency Level (% of Marks)	Targeted level of Attainment (% Students)
CO1	60	65
CO2	60	65
CO3	60	60
CO4	60	60
CO5	60	60
CO6	60	60

LESSON PLAN

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 1	Develop programs pointers	K3	8	Lecture	E-BOX
2	CO 2	Identify the type of function calls in C	K3	8	Lecture with Discussion	E-BOX
3	CO 3	Model the given problem statements using recursion functions in C	K3	8	Lecture with Discussion	E-BOX
4	CO 4	Construct programs using File Handling	K3	8	Lecture with Discussion	E-BOX
5	CO 5	Develop programs using Structures and Unions	K3	8	Lecture with Discussion	E-BOX
6	CO 6	Model the given problem statements using command line arguments and preprocessors to solve the given problems	K3	8	Lecture with Discussion	E-BOX

48

Total No. of Classes: 48